

HDOC DAY – 18

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer. Define one separate user-defined function for each menu option, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

The menu options are:

1. Check whether it is a strong number.
2. Check whether it is an Armstrong number.
3. Check whether it is a Harshad number.
4. Check whether it is a Duck number.

ALGORITHM:

1. Start.
2. Display the menu:
 - 1. Strong Number
 - 2. Armstrong Number
 - 3. Harshad Number
 - 4. Duck Number
3. Ask the user to enter a choice.
4. Use a switch-case statement.
5. If choice is 1, call strong Number ().
6. If choice is 2, call Armstrong Number ().
7. If choice is 3, call Harshad Number ().
8. If choice is 4, call duck Number ().
9. For any other choice, display "Invalid choice".
10. Stop.

>Algorithm for each function:

A. Strong Number

1. Start.
2. Enter a positive integer.
3. Store the original number.
4. Extract each digit using % 10.
5. Find the factorial of each digit.
6. Add all the factorials.
7. Compare the sum with the original number.
8. If equal, display “ Strong number.
9. Otherwise, display “Not a strong number .
10. Stop.

B. Armstrong Number

1. Enter a positive integer.
2. Count the number of digits.
3. Find each digit raised to the power of the number of digits.
4. Add these values.
5. If the sum equals the original number, display “Armstrong number”.
6. Otherwise, display “Not an Armstrong number.

C. Harshad Number

1. Enter a positive integer.
2. Find the sum of its digits.
3. Divide the original number by the sum.
4. If the remainder is 0, display “Harshad number”.
5. Otherwise, display “Not a Harshad number”.

D. Duck Number

1. Enter a positive integer.
2. Check each digit of the number.
3. If any digit is 0, display “Duck number”.
4. Otherwise, display “ Not a duck number”.

>Test Case Table:

Test Case	Menu Choice	Input	Expected Output
1	1	145	Strong Number
2	1	123	Not a Strong Number
3	2	153	Armstrong Number
4	2	123	Not an Armstrong Number
5	3	18	Harshad Number
6	3	19	Not a Harshad Number
7	4	102	Duck Number
8	4	123	Not a Duck Number

>Write a code C.

```
File Edit Search View Document Help
1 2 3 4
~/day11.c - Mousepad
Log Out...

1 #include <stdio.h>
2
3 void strongNumber();
4 void armstrongNumber();
5 void harshadNumber();
6 void duckNumber();
7
8 int main()
9 {
10     int choice;
11
12     printf("1. Strong Number\n");
13     printf("2. Armstrong Number\n");
14     printf("3. Harshad Number\n");
15     printf("4. Duck Number\n");
16
17     printf("Enter your choice: ");
18     scanf("%d", &choice);
19
20     switch(choice)
21     {
22         case 1:
23             strongNumber();
24             break;
25         case 2:
26             armstrongNumber();
27             break;
28         case 3:
29             harshadNumber();
30             break;
31         case 4:
32             duckNumber();
33             break;
34         default:
35             printf("Invalid Choice");
36     }
37
38     return 0;
39 }
40
41
42
43
44
45 void strongNumber()
46 {
47     int n, temp, digit, sum = 0, fact, i;
48
49     printf("Enter a number: ");
50     scanf("%d", &n);
51
52     temp = n;
```

```
File Edit Search View Document Help
temp = n;
while(temp > 0)
{
    digit = temp % 10;
    fact = 1;
    for(i = 1; i <= digit; i++)
        fact = fact * i;
    sum = sum + fact;
    temp = temp / 10;
}
if(sum == n)
    printf("Strong Number");
else
    printf("Not a Strong Number");
}
}

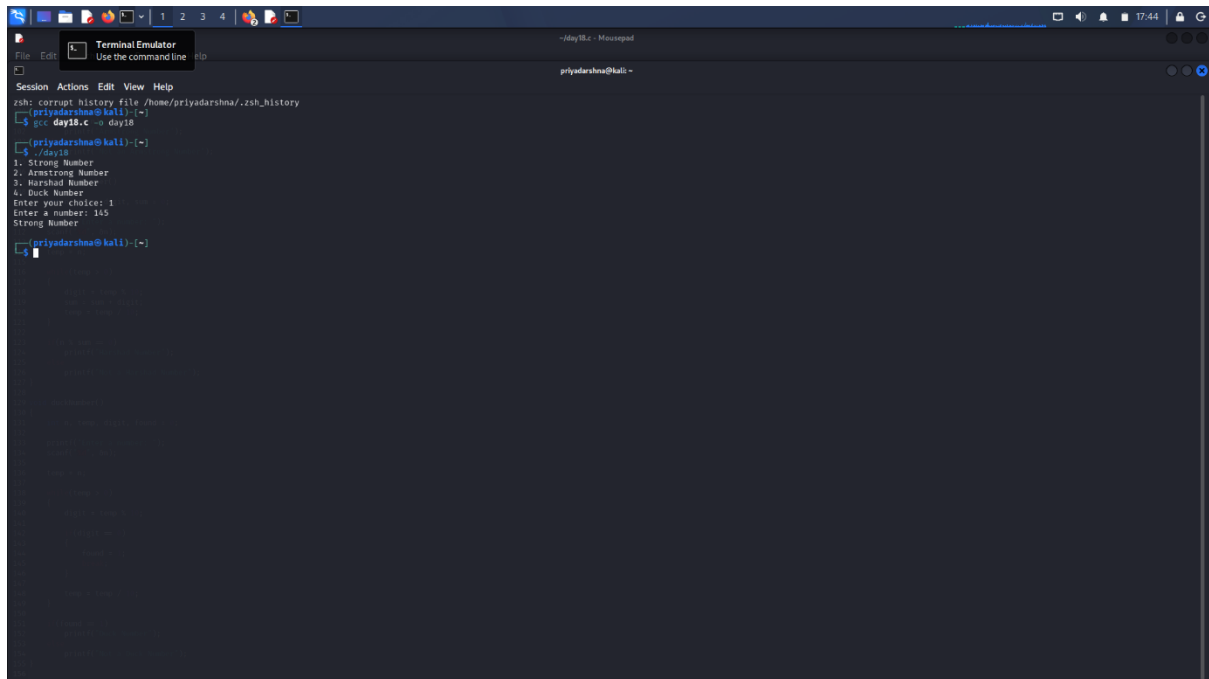
void armstrongNumber()
{
    int n, temp, digit, sum = 0, count = 0, i, power;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while(temp > 0)
    {
        count++;
        temp = temp / 10;
    }
    temp = n;
    while(temp > 0)
    {
        digit = temp % 10;
        power = 1;
        for(i = 1; i <= count; i++)
            power = power * digit;
        sum = sum + power;
        temp = temp / 10;
    }
    if(sum == n)
        printf("Armstrong Number");
    else
        printf("Not an Armstrong Number");
}
```

```
temp = temp / 10;
}
}
if(sum == n)
    printf("Armstrong Number");
else
    printf("Not an Armstrong Number");
}
}

void harshadNumber()
{
    int n, temp, digit, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while(temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit;
        temp = temp / 10;
    }
    if(n % sum == 0)
        printf("Harshad Number");
    else
        printf("Not a Harshad Number");
}

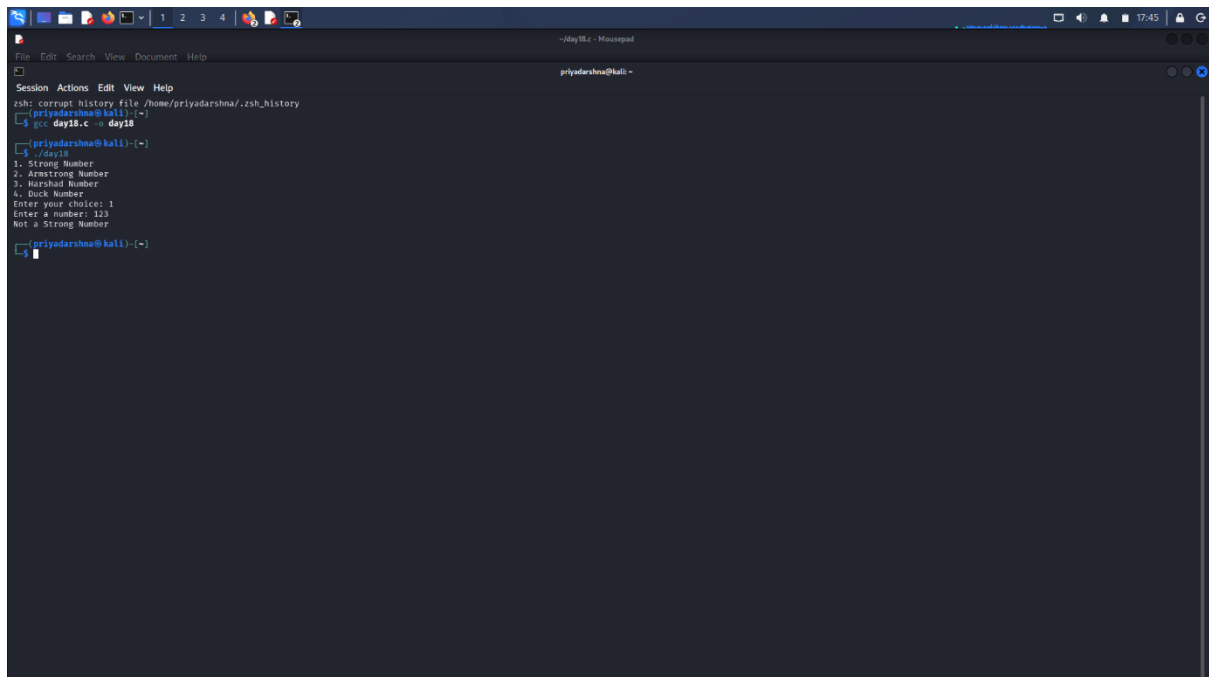
void duckNumber()
{
    int n, temp, digit, found = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit == 0)
        {
            found = 1;
            break;
        }
        temp = temp / 10;
    }
    if(found == 1)
        printf("Duck Number");
    else
        printf("Not a Duck Number");
}
```

>Test Case 1:



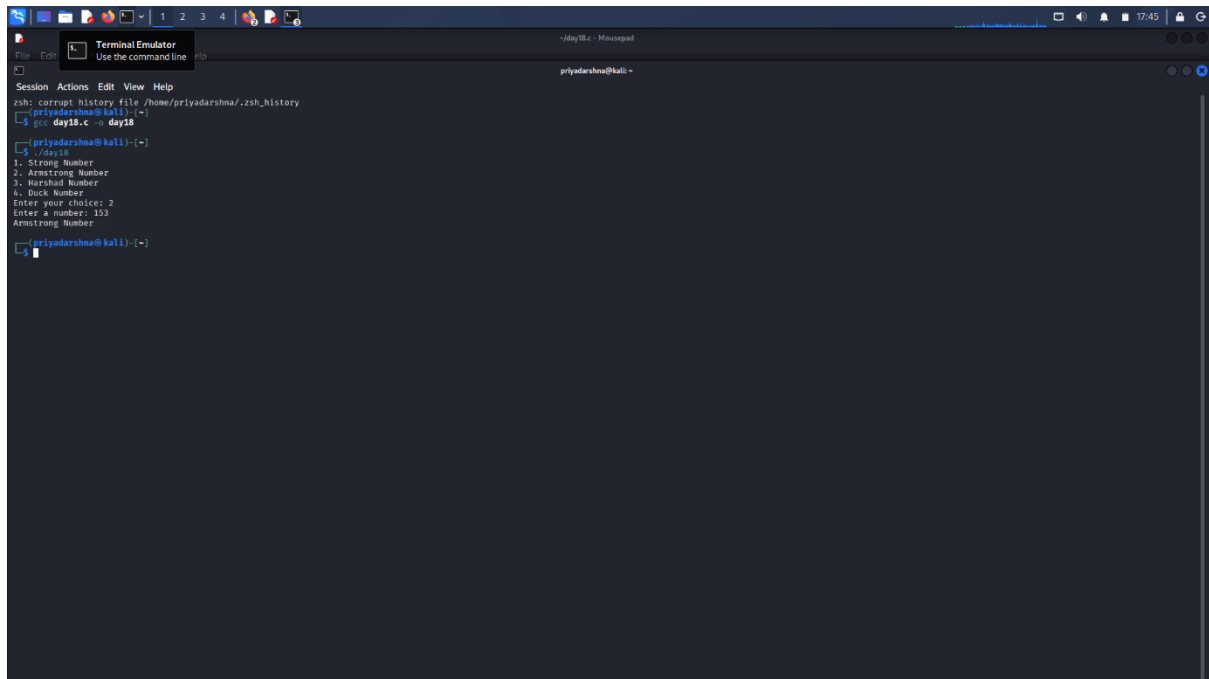
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day18.c -o day18
(priyadarshna@kali)~$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 1
Enter a number: 145
Strong Number
(priyadarshna@kali)~$
```

>Test Case 2:



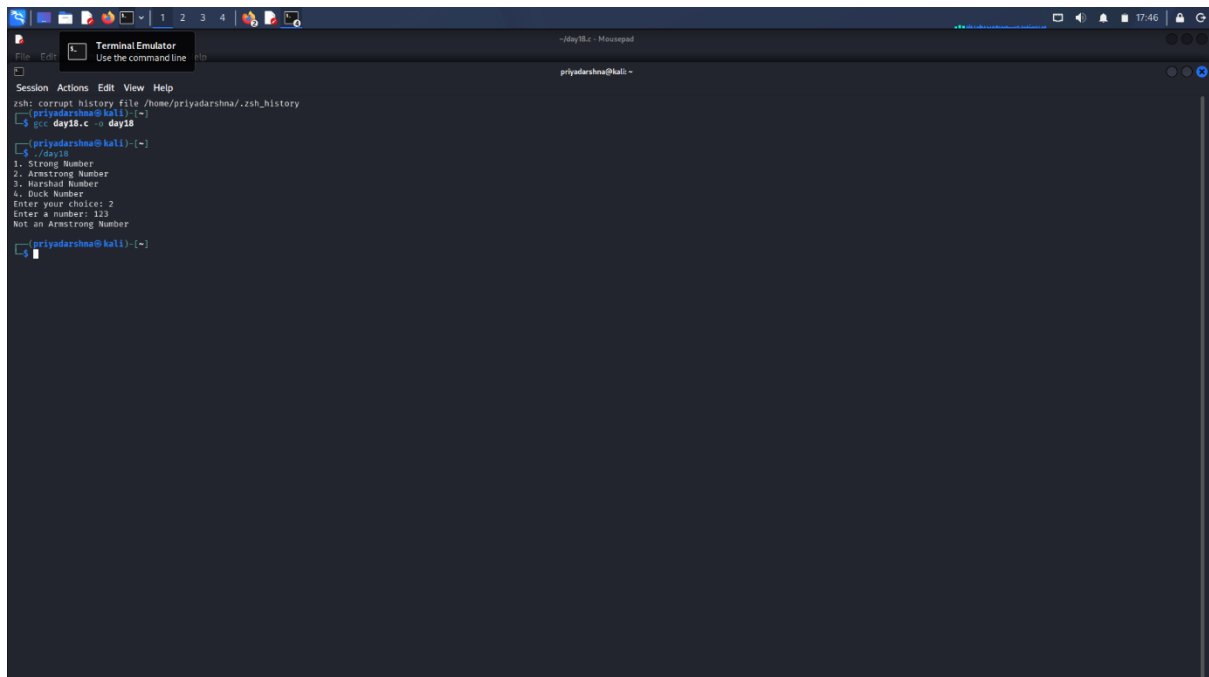
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali)~$ gcc day18.c -o day18
(priyadarshna@kali)~$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 1
Enter a number: 123
Not a Strong Number
(priyadarshna@kali)~$
```

>Test Case 3:



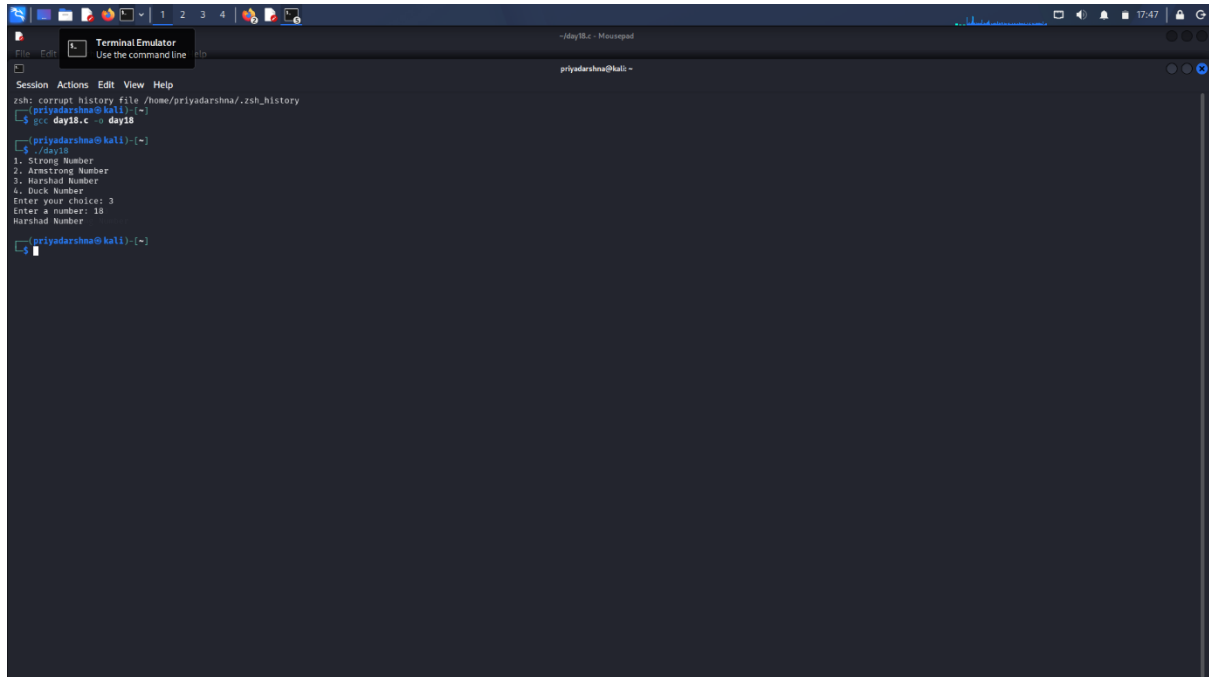
```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day18.c -o day18
(priyadarshna@kali) ~
$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 2
Enter a number: 153
Armstrong Number
(priyadarshna@kali) ~
$
```

>Test case 4:



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
(priyadarshna@kali) ~
$ gcc day18.c -o day18
(priyadarshna@kali) ~
$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 2
Enter a number: 123
Not an Armstrong Number
(priyadarshna@kali) ~
$
```

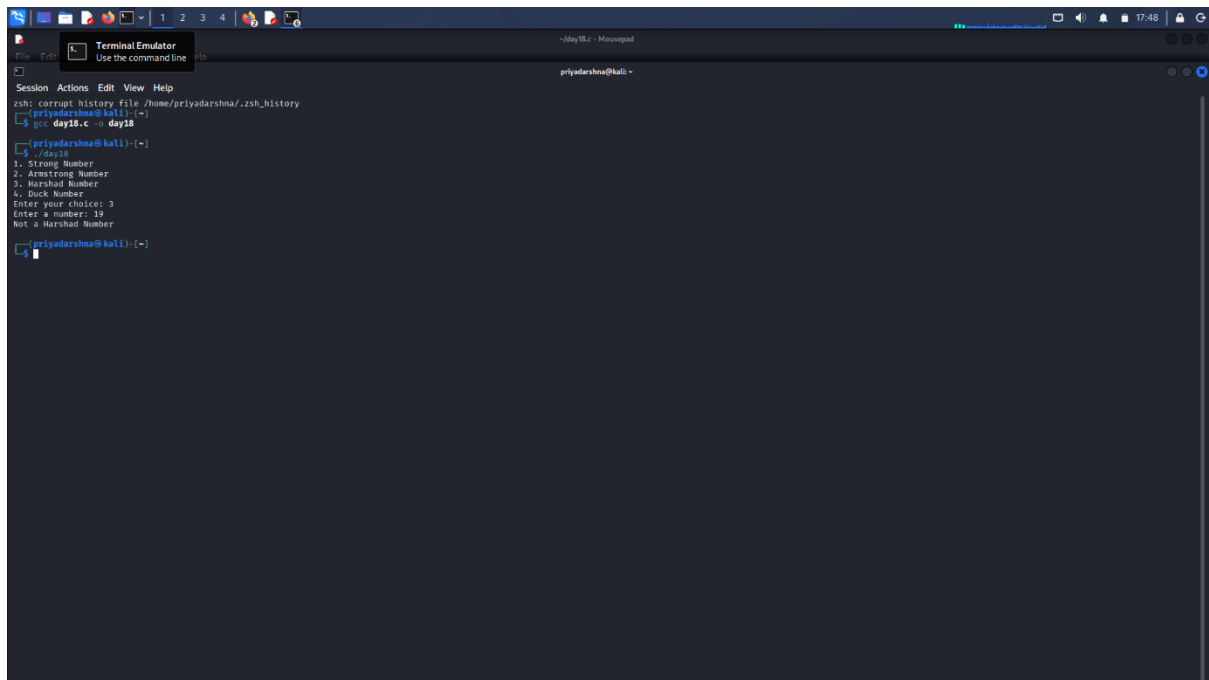
>Test case 5:



A screenshot of a terminal emulator window titled "Terminal Emulator" with the subtitle "Use the command line". The window shows a shell prompt "zsh: corrupt history file /home/priyadarshna/.zsh_history" and a user prompt "priyadarshna@kali: ~". The user enters the command "gcc day18.c -o day18". The prompt changes to "priyadarshna@kali: ~". The user enters the command "./day18". The program outputs a menu with four options: "1. Strong Number", "2. Armstrong Number", "3. Harshad Number", and "4. Duck Number". The user enters "3". The program prompts "Enter a number: 18". The user enters "18". The program outputs "Harshad Number".

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali: ~
$ gcc day18.c -o day18
priyadarshna@kali: ~
$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 3
Enter a number: 18
Harshad Number
priyadarshna@kali: ~
$
```

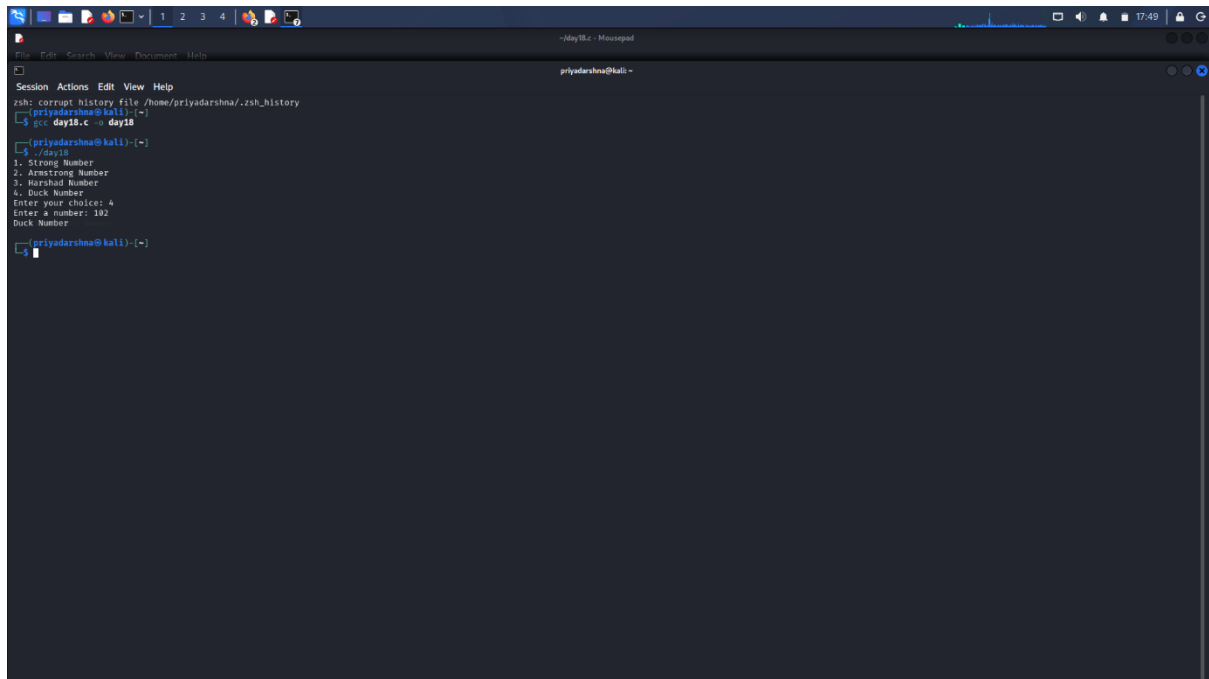
>Test Case 6:



A screenshot of a terminal emulator window titled "Terminal Emulator" with the subtitle "Use the command line". The window shows a shell prompt "zsh: corrupt history file /home/priyadarshna/.zsh_history" and a user prompt "priyadarshna@kali: ~". The user enters the command "gcc day18.c -o day18". The prompt changes to "priyadarshna@kali: ~". The user enters the command "./day18". The program outputs a menu with four options: "1. Strong Number", "2. Armstrong Number", "3. Harshad Number", and "4. Duck Number". The user enters "3". The program prompts "Enter a number: 19". The user enters "19". The program outputs "Not a Harshad Number".

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali: ~
$ gcc day18.c -o day18
priyadarshna@kali: ~
$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 3
Enter a number: 19
Not a Harshad Number
priyadarshna@kali: ~
$
```

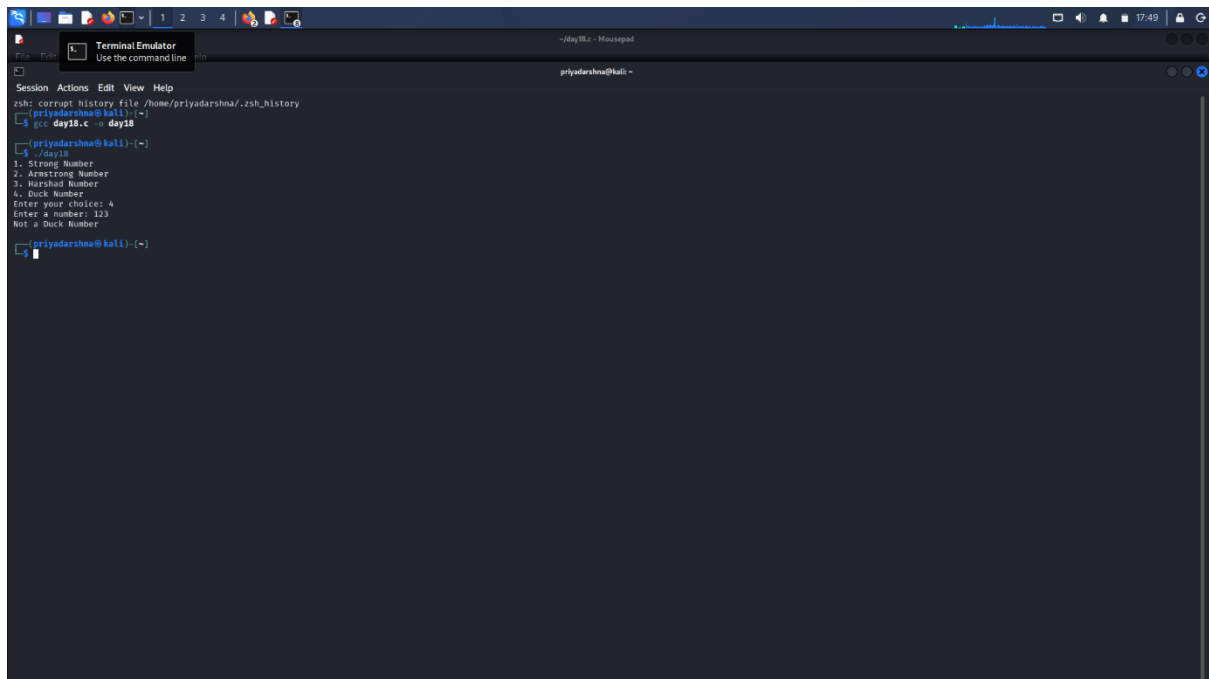
>Test Case 7:



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali:~`. The user enters `gcc day18.c -o day18` and then `./day18`. The program outputs a menu with four options: 1. Strong Number, 2. Armstrong Number, 3. Harshad Number, and 4. Duck Number. The user enters '4'. The program then prompts 'Enter your choice: 4' and 'Enter a number: 192'. The user enters '192'. The program outputs 'Duck Number'.

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali:~$ gcc day18.c -o day18
priyadarshna@kali:~$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 4
Enter a number: 192
Duck Number
priyadarshna@kali:~$
```

>Test Case 8:



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali:~`. The user enters `gcc day18.c -o day18` and then `./day18`. The program outputs a menu with four options: 1. Strong Number, 2. Armstrong Number, 3. Harshad Number, and 4. Duck Number. The user enters '4'. The program then prompts 'Enter your choice: 4' and 'Enter a number: 123'. The user enters '123'. The program outputs 'Not a Duck Number'.

```
zsh: corrupt history file /home/priyadarshna/.zsh_history
priyadarshna@kali:~$ gcc day18.c -o day18
priyadarshna@kali:~$ ./day18
1. Strong Number
2. Armstrong Number
3. Harshad Number
4. Duck Number
Enter your choice: 4
Enter a number: 123
Not a Duck Number
priyadarshna@kali:~$
```